Higher National Diploma in Quantity Surveying
2nd Year, 2nd Semester Examination – 2020
QS 22023 Building Economics II

Instructions for Candidates:
Answer five questions only
All Questions carry equal marks

No. of questions: 06
No. of pages : 03
Time : **03 hours**

Question 01

I. What are the purposes of Approximate Estimation? (3 Marks)

II. What are the preliminary estimation methods available? (list out at least four methods) (4 Marks)

III. What are the factors that influence the selection of the most appropriate preliminary estimation methods? (list out at least four factors) (4 Marks)

IV. What are the adjustments need to be done to arrive a rate from a previously completed similar project? (list out at least four adjustments) (4 Marks)

V. What are the information / documents needed to do a preliminary estimation? (list out at least five documents) (5 Marks)

(Total 20 Marks)

Question 02

You are working as Senior Quantity Surveyor / Cost Manager for a leading Cost Consultant. Government Department of Sri Lanka approached you to get some cost advise for build schools in rural area.

I. How will you prepare the preliminary estimate for above mentioned School project with very minimal design information? (only few sketch available) (4 Marks)

II. Propose the most appropriate preliminary estimation method for the above mentioned school project at the appraisal stage (RIBA 2007) of the project? (3 Marks)

III. Provide your justification for above selection? (Question No. II) (5 Marks)

IV. You have a previously (long time ago) completed similar type of project (similar size, design and specification) cost data but, it is far away from proposed new project, how you will adjust the previously completed project rate to proposed new project? (4 Marks)

V. You are requested to prepare the preliminary estimate for a School and it needs to be accommodated 650 students. You have recently completed similar project cost data for Rs 25,000,000.00 and it accommodates 550 students at same district. What is your

estimated cost for the proposed school with the available information? List out if you made any assumptions? (4 Marks)

(Total 20 Marks)

Question 03

I. What are the key purposes of cost planning? (3 Marks)

II. Briefly describe the process of cost planning process? (5 Marks)

III. What are the information required for a cost plan? (4 Marks)

IV. What are the core components of a cost plan layout? (4 Marks)

V. What are the sources of cost data for a cost plan? (4 Marks)

(Total 20 Marks)

Question 04

I. Labour, material, plant, overhead and profit are basic elements of unit rate. What are the other considerations that need to be included in the unit rate to establish a more reliable unit rate? (4 Marks)

II. What are the basic details required to build a Labour rate for permanent works? (4 Marks)

III. You have built a unit rate for suspended ceiling works but when compared with market conditions, it is lower / higher than the market rate. What are the reasons for the lower / higher rate? (4 Marks)

IV. What are the factors you will consider while taking the decision to purchase or hire a plant for a construction company? (4 Marks)

V. You have been appointed as head of the Tendering Department in a Contracting Organization. You need to build an all in Material unit rate for your ongoing tendering projects. What are the things you need to consider? (4 Marks)

(Total 20 Marks)

Question 05

I. What are the design factors influencing the cost of the project? (List out at least five major factors) (5 Marks)

II. What are the costs included in the life cycle cost (LCC)? (3 Marks)

III. List out the major roles of Quantity Surveyor in following stages of the project?

Appraisal

Technical Design

Production Information

Tender Action

Construction to practical completion (4 Marks)

IV. Briefly discuss about the pre-tender estimate / Engineer's estimate? (4 Marks)

V. Briefly discuss about the detailed estimate? (4 Marks)

(Total 20 Marks)

Question 06

I. Define project network techniques? (4 Marks)

II. What is activity and event? (3Marks)

III. Names commonly used two types of network system? (3Marks)

IV. Draw the network for this project? Prepare an activity schedule showing EST, EFT, and LST, LFT, and total float and critical path? (10 Marks)

Activity	Preceding activity	Duration weeks
A	-	7
B	A	4
C	A	8
D	B	14
E	D	12
F	D	11
G	D	7
H	B	11
I	C,E	3
J	G	4
K	F,I,J	5
L	K	11
M	H,G	9
N	M	11

(Total 20 Marks)